Lecture 2

rotation, Lie / quaternion, SE(3), $SE_2(3)$

The basic math operation:

$$e^x = \exp(x) = \sum_{k=0}^{\infty} \frac{1}{k!} x^k = I + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \frac{1}{4!}x^3 + \cdots\cdots$$

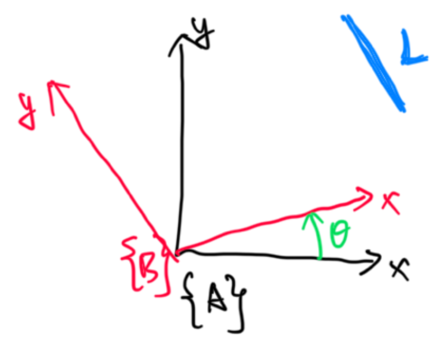
$$\sin(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1} = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - + \cdots\cdots$$

$$\cos(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k} = I - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \cdots\cdots$$

If $x \to 0$, $x^2 \to 0$, $x^3 \to 0$, then $e^x \approx I + x$, $\sin(x) \approx x$, $\cos(x) = I$

These are the important approximation.

Notations:



$\{A\}$: frame $\{A\}$

$\{B\}$: frame $\{B\}$

^{A}L: L represented in $\{A\}$

^{B}L: L represented in $\{B\}$

$\theta_{A\to B}$: rotation angle aroud axis z: $\{A\} \to \{B\}$

Hence: we define $^{A}_{B}R$ as:

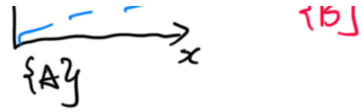
$$^{A}L = {}^{A}_{B}R \cdot {}^{B}L$$

$$\hat{=}\ {}^{A}_{B}R(\theta_{A\to B})\, {}^{B}L$$

we define translation as:

$^{A}x \hat{=}\ ^{A}B$

p_B - $V_{A\to B}$



overall speaking, the full transformation:

$$\{A\},\{B\} \Rightarrow \{{}^A_BR, {}^Ap_B\} \Rightarrow {}^A_BT = \begin{bmatrix} {}^A_BR & {}^Ap_B \\ 0_{1\times3} & 1 \end{bmatrix}$$

$$\boxed{\begin{aligned} &R = [v_1\ v_2\ v_3] \quad RR^T = R^TR = I,\ R^T = R^{-1} \\ &v_3 = v_1\times v_2,\quad v_2 = v_3\times v_1,\quad v_1 = v_2\times v_3 \end{aligned}}$$

How to define rotation?

roll: θ (x) pitch: γ (y) yaw: ψ (z)

$$R(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix} \quad R(\gamma) = \begin{bmatrix} \cos\gamma & 0 & \sin\gamma \\ 0 & 1 & 0 \\ -\sin\gamma & 0 & \cos\gamma \end{bmatrix} \quad R(\psi) = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R = R(\psi)R(\gamma)R(\theta) = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos\gamma & 0 & \sin\gamma \\ 0 & 1 & 0 \\ -\sin\gamma & 0 & \cos\gamma \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

yaw × pitch × roll

what if $\delta\theta\to0, \delta\gamma\to0, \delta\psi\to0$, then:

$$R(\delta\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -\theta \\ 0 & \theta & 1 \end{bmatrix} \quad R(\delta\gamma) = \begin{bmatrix} 1 & 0 & \gamma \\ 0 & 1 & 0 \\ -\gamma & 0 & 1 \end{bmatrix} \quad R(\delta\psi) = \begin{bmatrix} 1 & -\psi & 0 \\ \psi & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R = R(\delta\psi)R(\delta\gamma)R(\delta\theta)$$

$$= \begin{bmatrix} 1 & -\delta\psi & 0 \\ \delta\psi & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & \delta\gamma \\ 0 & 1 & 0 \\ -\delta\gamma & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -\delta\theta \\ 0 & \delta\theta & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -\delta\psi & \delta\gamma \\ \delta\psi & 1 & \underset{0}{\delta\psi\gamma} \\ -\delta\gamma & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -\delta\theta \\ 0 & \delta\theta & 1 \end{bmatrix} = \begin{bmatrix} 1 & -\delta\psi & \underset{0}{\delta\psi\theta}+\delta\gamma \\ \delta\psi & 1 & -\delta\theta \\ -\delta\gamma & \delta\theta & 1 \end{bmatrix}$$

$$= I + \begin{bmatrix} 0 & -\delta\psi & \delta\gamma \\ \delta\psi & 0 & -\delta\theta \\ -\delta\gamma & \delta\theta & 0 \end{bmatrix} \qquad \theta = \begin{bmatrix} \delta\theta \\ \delta\gamma \\ \delta\psi \end{bmatrix}$$

$$\begin{bmatrix} \delta\theta \\ \delta r \\ \delta\varphi \end{bmatrix}_\times = \begin{bmatrix} 0 & -\delta\varphi & \delta r \\ \delta\varphi & 0 & -\delta\theta \\ -\delta r & \delta\theta & 0 \end{bmatrix} \rightarrow \text{skew-symetric matrix}$$

$$\begin{bmatrix} \delta\theta \\ \delta r \\ \delta\varphi \end{bmatrix}_\times = - \begin{bmatrix} \delta\theta \\ \delta r \\ \delta\varphi \end{bmatrix}_\times^T$$

$$R(\theta) = R(\delta\varphi)\, R(\delta r)\, R(\delta\theta) \approx I + \lfloor \theta \rfloor_\times \approx \exp(\lfloor \theta \rfloor_\times)$$

Let's define $\mathrm{Exp}(\theta) \triangleq \exp(\lfloor \theta \rfloor_\times)$,

should we suppose: $R(\theta) = \mathrm{Exp}(\theta)$?

Axis-angle rotation

k: rotation axis, θ: rotation angle:

$$k = \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix}, \quad \lfloor k \rfloor_\times = \begin{bmatrix} 0 & -k_3 & k_2 \\ k_3 & 0 & -k_1 \\ -k_2 & k_1 & 0 \end{bmatrix}$$

$$\theta = \theta k = \begin{bmatrix} \theta k_1 \\ \theta k_2 \\ \theta k_3 \end{bmatrix} = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix} \qquad \lfloor \theta \rfloor_\times = \begin{bmatrix} 0 & -\theta_3 & \theta_2 \\ \theta_3 & 0 & -\theta_1 \\ -\theta_2 & \theta_1 & 0 \end{bmatrix}$$

$$\lfloor k \rfloor_\times^2 = k k^T - I_3 \qquad \lfloor k \rfloor_\times^5 = \lfloor k \rfloor_\times \qquad \lfloor k \rfloor_\times = -\lfloor k \rfloor_\times^T$$

$$\lfloor k \rfloor_\times^3 = -\lfloor k \rfloor_\times \qquad \lfloor k \rfloor_\times^6 = \lfloor k \rfloor_\times^2$$

$$\lfloor k \rfloor_\times^4 = -\lfloor k \rfloor_\times^2 \qquad \lfloor k \rfloor_\times^7 = -\lfloor k \rfloor_\times$$

Hence:

$$\begin{aligned}
\mathrm{Exp}(\theta) &= \exp(\theta \lfloor k \rfloor_\times) \\
&= I + \theta \lfloor k \rfloor_\times + \frac{\theta^2}{2!}\lfloor k \rfloor_\times^2 + \frac{\theta^3}{3!}\lfloor k \rfloor_\times^3 + \frac{\theta^4}{4!}\lfloor k \rfloor_\times^4 + \frac{\theta^5}{5!}\lfloor k \rfloor_\times^5 + \frac{\theta^6}{6!}\lfloor k \rfloor_\times^6 + \frac{\theta^7}{7!}\lfloor k \rfloor_\times^7 + \\
&= I + \theta \lfloor k \rfloor_\times + \frac{\theta^2}{2!}\lfloor k \rfloor_\times^2 - \frac{\theta^3}{3!}\lfloor k \rfloor_\times - \frac{\theta^4}{4!}\lfloor k \rfloor_\times^2 + \frac{\theta^5}{5!}\lfloor k \rfloor_\times + \frac{\theta^6}{6!}\lfloor k \rfloor_\times^2 - \frac{\theta^7}{7!}\lfloor k \rfloor_\times - \cdots \\
&= I + \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \cdots\right)\lfloor k \rfloor_\times + \left(\frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \frac{\theta^6}{6!} - \cdots\right)\lfloor k \rfloor_\times^2
\end{aligned}$$

$$= I + \underbrace{\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \cdots\right)}_{\sin\theta} \lfloor k \rfloor_\times + \underbrace{\left(\frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \frac{\theta^6}{6!} - \cdots\right)}_{1-\cos\theta} \lfloor k \rfloor_\times^2$$

$$= I + \sin\theta \lfloor k \rfloor_\times + (1-\cos\theta) \lfloor k \rfloor_\times^2 \Rightarrow \text{Rodrigues' rotation equation}$$

$$R(\theta) = \mathrm{Exp}(\theta) = \exp(\theta \lfloor k \rfloor_\times) = I + \sin\theta \lfloor k \rfloor_\times + (1-\cos\theta) \lfloor k \rfloor_\times^2$$

If $\delta\theta \to 0$, $R(\delta\theta) \approx I + \lfloor \delta\theta \rfloor_\times$, small angle approximations

we should also define logarithmic maps:

Logarithmic map is the inverse of the exponential map:

$$\begin{cases} \log(R) = \lfloor \theta \rfloor_\times \\ \mathrm{Log}(R) = \theta_{3\times1} \end{cases} \qquad \begin{cases} \theta = \arccos\left(\dfrac{\mathrm{trace}R - 1}{2}\right) \\ \lfloor k \rfloor_\times = \dfrac{(R - R^T)}{2\sin\theta} \end{cases}$$

A few important operations:

$$\underset{\mathbb{R}^3}{\theta_{3\times1}} \to \underset{\text{Lie algra}}{\lfloor \theta \rfloor_{\times\,3\times3}} \to \underset{\text{Lie Group}}{\begin{matrix} R = \exp(\lfloor \theta \rfloor_\times) \\ R = \mathrm{Exp}(\theta) \end{matrix}} \quad \Bigg| \quad R_{3\times3} \to \lfloor \theta \rfloor_{\times\,3\times3} \to \begin{matrix} \theta_{3\times1} = \mathrm{Log}(R) \\ \lfloor \theta \rfloor_\times = \log(R) \end{matrix}$$

① $RR^T = I_3 = R^TR$ $\quad \frac{d}{dt}(RR^T) = 0 \Rightarrow \dot{R}R^T + R\dot{R}^T = 0$

$\dot{R}R^T = -R\dot{R}^T = -(\dot{R}R^T)^T \Rightarrow \dot{R}R^T$ should be skew-symetric

Let the skew-symetric matrix: $\begin{bmatrix} 0 & -w_3 & w_2 \\ w_3 & 0 & -w_1 \\ -w_2 & w_1 & 0 \end{bmatrix} = \lfloor w \rfloor_\times$

$\dot{R}R^T = \lfloor w \rfloor_\times \Rightarrow \dot{R} = R\lfloor w \rfloor_\times$

② $R(t)$, $R(t+\Delta t) = R(t) \cdot \exp(w\Delta t)$

$$\dot{R}(t) = \lim_{\Delta t \to 0} \frac{R(t+\Delta t) - R(t)}{\Delta t} = \lim_{\Delta t \to 0} \frac{R(t)\exp(w\Delta t) - R(t)}{\Delta t}$$

$$= \lim_{\Delta t \to 0} \frac{R(t)\exp(w\Delta t)}{\Delta t} = R(t) \lim_{\Delta t\to 0} \frac{\exp(w\Delta t)}{\Delta t} = R(t)\lfloor w \rfloor$$

③ Integration: $R_1 = \exp(w_1 \cdot \Delta t_1)$

$$R_{1\to k} = R_k \cdot R_{k-1} \cdots R_2 \cdot R_1 = \exp(w_k \Delta t_k) \cdots \exp(w_1 \cdot \Delta t_1)$$

Interpolation

④ ${}^{G}_{t_1}R$, ${}^{G}_{t_2}R$, ${}^{t_1}_{t_2}R = {}^{G}_{t_1}R^{T}\,{}^{G}_{t_2}R$, $t \in [t_1, t_2]$

$${}^{t_1}_{t_2}\theta = \log({}^{t_1}_{t_2}R) = \log({}^{G}_{t_1}R^{T}\,{}^{G}_{t_2}R) \qquad \lambda = \frac{t-t_1}{t_2-t_1}$$

$${}^{G}_{t}R = {}^{G}_{t_1}R \cdot {}^{t_1}_{t}R = {}^{G}_{t_1}R \cdot \exp\left(\frac{t-t_1}{t_2-t_1}\,{}^{t_1}_{t_2}\theta\right)$$

⑤ Given Rotation $R(\theta) = \mathrm{Exp}(\theta)$,

small angle in R^3: $\delta\theta$

small angle in $SO(3)$: $\delta\varphi$

$$R(\theta)\cdot \mathrm{Exp}(\delta\varphi) = R(\theta+\delta\theta)$$

$$\mathrm{Exp}(\theta)\,\mathrm{Exp}(\delta\varphi) = \mathrm{Exp}(\theta+\delta\theta)$$

$$\mathrm{Exp}(\delta\varphi) = \mathrm{Exp}(\theta)^{-1}\,\mathrm{Exp}(\theta+\delta\theta)$$

$$\delta\varphi = \mathrm{Log}\left(\mathrm{Exp}(\theta)^{-1}\,\mathrm{Exp}(\theta+\delta\theta)\right)$$

$$\frac{\partial \delta\varphi}{\partial \delta\theta} = \lim_{\delta\theta\to 0} \frac{\mathrm{Log}\left(\mathrm{Exp}(\theta)^{-1}\,\mathrm{Exp}(\theta+\delta\theta)\right)}{\delta\theta} \triangleq J_r(\theta)$$

$$\mathrm{Exp}(\theta+\delta\theta) = \mathrm{Exp}(\theta)\,\mathrm{Exp}(J_r(\theta)\cdot\delta\theta)$$

$$\mathrm{Exp}(\theta)\,\mathrm{Exp}(\delta\varphi) = \mathrm{Exp}(\theta + J_r^{-1}(\theta)\,\delta\varphi)$$

$$\Rightarrow \mathrm{Log}\left(\mathrm{Exp}(\theta)\,\mathrm{Exp}(\delta\varphi)\right) = \theta + J_r^{-1}(\theta)\,\delta\varphi$$

with $J_r(\theta) = I - \frac{1-\cos\|\theta\|}{\|\theta\|^2} \lfloor\theta\rfloor_\times + \frac{\|\theta\| - \sin\|\theta\|}{\|\theta\|^3} \lfloor\theta\rfloor_\times^2$

(6) $\mathrm{Exp}(\theta+\delta\theta) = \mathrm{Exp}(\theta)\,\mathrm{Exp}(J_r(\theta)\,\delta\theta)$

$= \mathrm{Exp}(J_l(\theta)\,\delta\theta)\,\mathrm{Exp}(\theta)$

$\Rightarrow J_l(\theta) = \mathrm{Exp}(\theta)\,J_r(\theta) = R(\theta)\,J_r(\theta)$

(7) $\mathrm{Exp}(\theta+\delta\theta) = \mathrm{Exp}(\theta)\,\mathrm{Exp}(J_r(\theta)\,\delta\theta)$

$\mathrm{Exp}(-\theta-\delta\theta) = \mathrm{Exp}(-J_l(-\theta)\,\delta\theta)\,\mathrm{Exp}(-\theta)$

$= \mathrm{Exp}(-J_r(\theta)\,\delta\theta)\,\mathrm{Exp}(-\theta)$

Hence: $J_l(-\theta) = J_r(\theta)$

(8) From (5) (6) (7):
$$\begin{cases} \mathrm{Exp}(\theta+\delta\theta) = \mathrm{Exp}(\theta)\,\mathrm{Exp}(J_r(\theta)\,\delta\theta) \\ J_l(\theta) = R(\theta)\,J_r(\theta) \\ J_l(-\theta) = J_r(\theta) \end{cases}$$

(9) Important properties: $\lfloor Rw\rfloor_\times = R\lfloor w\rfloor_\times R^T$:

Homework: How to prove it:?

$\lfloor Ru\rfloor_\times RV = R(\lfloor u\rfloor_\times V) = R(\lfloor u_\times\rfloor R^T RV)$

$= R\lfloor u_\times\rfloor R^T \cdot RV$

This is true for any RV, hence: $\lfloor Ru\rfloor_\times = R\lfloor u_\times\rfloor R^T$

(10): $\mathrm{Exp}(R\varphi) = R\,\mathrm{Exp}(\varphi)\,R^T$

Homework: prove this?

$$Exp(R\varphi) = I + \lfloor R\varphi \rfloor_\times + \frac{1}{2!}\lfloor R\varphi \rfloor_\times^2 + \frac{1}{3!}\lfloor R\varphi \rfloor_\times^3 + \cdots$$

$$\lfloor R\varphi \rfloor_\times^k = R\lfloor \varphi \rfloor_\times R^T \cdots\cdots R\lfloor \varphi \rfloor_\times R^T = R\lfloor \varphi \rfloor_\times^k R^T$$

$$Exp(R\varphi) = R\left(I + \lfloor \varphi \rfloor_\times + \frac{1}{2!}\lfloor \varphi \rfloor_\times^2 + \frac{1}{3!}\lfloor \varphi \rfloor_\times^3 + \cdots\right)R^T$$

$$= R\, Exp(\varphi) R^T$$

(11) Given $\{C\}$, $\{I\}$ camera frame, Imu frame, ${}^C_I R$ rotation from I to C

${}^I v$, ${}^C v$ represents vector v in $\{I\}$ frame and $\{C\}$ frame

$$\begin{cases} {}^C v_1 = {}^C_I R\, {}^I v_1 & (1) \\ {}^C v_2 = {}^C_I R\, {}^I v_2 & (2) \end{cases}$$

How to use (1) and (2) to compute ${}^C_I R$?

The trick: $[{}^C v_1 \;\; {}^C v_2 \;\; {}^C v_1 \times {}^C v_2] = {}^C_I R [{}^I v_1 \;\; {}^I v_2 \;\; {}^I v_1 \times {}^I v_2]$

(12) what if given:

$$\begin{cases} {}^{C_1}_{C_2} R = {}^C_I R\, {}^{I_1}_{I_2} R\, {}^C_I R^T & (3) \\ {}^{C_2}_{C_3} R = {}^C_I R\, {}^{I_2}_{I_3} R\, {}^C_I R^T & (4) \end{cases}$$

How to use (3) and (4) to compute ${}^C_I R$?

(13) Discussion: (11) and (12):

(1) what's the minnum pair of data needed?

(2) what's the requirement of these data pairs?

(3) what if we have more than minimum pair of data?

(14) Given n measurents R_{m_i} of R with Gaussian noise n_i, $i = 1, \cdots n$,

$$R_{m_i} = R \cdot exp(n_i) \quad n_i \sim N(0_{3\times1}, P_i) \quad i = 1, \cdots, n$$

How to evaluate R and its covariance?

The above is rotation with SO(3). Now let's go to SE(3)

SO(3): special orthogonal group

$$SO(3) = \{ R \in R^{3\times3} \mid R^T R = I_3,\ \det(R) = 1 \}$$

4 properties: closure $R_1 R_2 \in SO(3)$

identity: $I_3 \in SO(3)$

Inverse: $R^{-1} = R^T \in SO(3)$

Associativity: $(R_1 R_2) R_3 = R_1 (R_2 R_3)$ for $R_i \in SO(3)$

SE(3): special Euclidean group:

$$SE(3) = \left\{ T = \begin{bmatrix} R & p \\ 0_{1\times3} & 1 \end{bmatrix}_{4\times4} \middle|\ R \in SO(3),\ p \in R^3 \right\}$$

4 properties: closure: $T_1 \cdot T_2 = \begin{bmatrix} R_1 & p_1 \\ 0_{1\times3} & 1 \end{bmatrix} \begin{bmatrix} R_2 & p_2 \\ 0_{1\times3} & 1 \end{bmatrix} = \begin{bmatrix} R_1 R_2 & R_1 p_2 + p_1 \\ 0_{1\times3} & 1 \end{bmatrix}$

Identity: $\begin{bmatrix} I_3 & 0_{3\times1} \\ 0_{1\times3} & 1 \end{bmatrix} \in SE(3)$

Inverse: $\begin{bmatrix} R & p \\ 0_{1\times3} & 1 \end{bmatrix}^{-1} = \begin{bmatrix} R^T & -R^T p \\ 0_{1\times3} & 1 \end{bmatrix} \in SE(3)$

Associativity: $(T_1 T_2) T_3 = T_1 (T_2 T_3)$ for all $T_1, T_2, T_3 \in SE(3)$

Matrix Lie Group:

① A group is a set of elements with an operation that combines any two elements to form a third one also in the set. A group satisfies: closure, associativity, identity and invertibility.

② A Lie Group is a group that is also differentiable manifold with the property that the group operation are smooth.

that the group ope...

③ A matrix Lie group: the Lie group elements are matrices.

Let's introduce the operations between different spaces:

vector space	$\longrightarrow$	Lie Algebra	$\longrightarrow$	Lie Group
R^m		$so(n)$		$SO(n)$
x	$\wedge$: wedge opreation	$x^\wedge$	exponential	$\exp(x^\wedge)$

vector space	$\longleftarrow$	Lie Algebra	$\longleftarrow$	Lie Group
$\log(X)^\vee$	$\vee$: Vee operation	$\log(X)$	logrithmic	X

For simplicity, $\begin{cases} X = \mathrm{Exp}(x) = \exp(x^\wedge) \\ x = \mathrm{Log}(X) = \log(X)^\vee \end{cases}$

Now: let's use $SO(3)$: for example:

$\theta_{3\times1} \in R^3 \rightarrow$ Lie Algebra: $\lfloor\theta\rfloor_\times \triangleq \theta^\wedge \rightarrow$ Lie Group: $R = \exp(\lfloor\theta\rfloor_\times)$
$= \exp(\theta^\wedge)$

$R \in SO(3) \rightarrow$ Lie Algebra: $\lfloor\theta\rfloor_\times \triangleq \theta^\wedge = \log(R) \rightarrow R^3$: $\theta = \log(R)^\vee$

For the properties of $SO(3)$, please refer to the rotation part.

Add one more discussion about the left and right Jacobians:

$$R(\theta) = \mathrm{Exp}(\theta) = I_3 + \lfloor\theta\rfloor_\times + \frac{1}{2!}\lfloor\theta\rfloor_\times^2 + \frac{1}{3!}\lfloor\theta\rfloor_\times^3 + \cdots$$
$$= I_3 + \lfloor\theta\rfloor_\times J_L(\theta) \triangleq I_3 + \theta^\wedge J_L(\theta)$$

with $J_L(\theta) = \sum_{n=0}^{\infty} \frac{1}{(n+1)!}(\theta^\wedge)^n$

Prove: ① $J_L(\theta) = I_3 + \left(\frac{1-\cos\|\theta\|}{\|\theta\|^2}\right)\theta^\wedge + \left(\frac{\|\theta\| - \|\sin\theta\|}{\|\theta\|^3}\right)(\theta^\wedge)^2$

② if $\theta \to 0$, $J_L(\theta) \to$?

② 若 $\theta \to 0$, $J_l(\theta) \to$?

③ $\mathrm{Exp}(\theta + \delta\theta) \approx \mathrm{Exp}(J_l(\theta)\delta\theta)\,\mathrm{Exp}(\theta)$

Thought: why we need these complicated math?

① It is actually simplifies the problem a lot. It actually ~~make the math~~ simpler

③ The core idea: use smooth manifold to model the unsmoothed parameters in vector space, which makes the optimization, interpolation, integration, derivative easier even possible.

For example: $\theta \to [-\pi, \pi]$, has cut off for angles.

The arc of the sphere is smooth, easy for any operation

The red line is the tangent line: which can be used to model the derivative of the arc, the small change/error states for optimisation.

Now: Let's begin the SE(3)

Vector space $\longrightarrow$ Lie algebra $\longrightarrow$ Lie Group.

$$\xi = \begin{bmatrix} \theta \\ \rho \end{bmatrix} \in R^6 \qquad \xi^\wedge = \begin{bmatrix} \theta^\wedge & \rho \\ 0_{1\times3} & 1 \end{bmatrix} = \begin{bmatrix} \lfloor\theta\rfloor_\times & \rho \\ 0_{1\times3} & 1 \end{bmatrix}$$

$$T = \exp\left(\begin{bmatrix} \theta^\wedge & \rho \\ 0_{1\times3} & 1 \end{bmatrix}\right) = \begin{bmatrix} \exp(\lfloor\theta\rfloor_\times) & J_l(\theta)\rho \\ 0_{1\times3} & 1 \end{bmatrix} = \begin{bmatrix} R & p \\ 0_{1\times3} & 1 \end{bmatrix}$$

Then

vector space $\longleftarrow$ Lie algebra $\longleftarrow$ Lie Group

$$T = \begin{bmatrix} R & p \\ 0_{1\times3} & 1 \end{bmatrix}$$

$$\log(T)^\vee = \begin{bmatrix} \theta \\ \rho \end{bmatrix} = \xi$$

$$\begin{aligned} \xi^\wedge &= \log(T) \\ &= \log\left(\begin{bmatrix} R & p \\ 0_{1\times3} & 1 \end{bmatrix}\right) \\ &= \begin{bmatrix} \log(R) & J_L^{-1}(\theta)\, p \\ 0_{1\times3} & 1 \end{bmatrix} \\ &= \begin{bmatrix} \theta & \rho \\ 0_{1\times3} & 1 \end{bmatrix} \end{aligned}$$

Discussion: why: $\exp\left(\begin{bmatrix} \log R & \rho \\ 0_{1\times3} & 1 \end{bmatrix}\right) = \begin{bmatrix} R & J_L(\theta)\,\rho \\ 0_{1\times3} & 1 \end{bmatrix}$? prove it!

we still define : $\begin{cases} \mathrm{Exp}(\xi) = T \\ \mathrm{Log}(T) = \xi \end{cases}$

Left and Right Jacobians:

$$\begin{aligned} \mathrm{Exp}(\xi + \delta\xi) &= \mathrm{Exp}(\xi)\cdot \mathrm{Exp}(J_R(\xi)\,\delta\xi) \\ &= \mathrm{Exp}(J_L(\xi)\,\delta\xi)\,\mathrm{Exp}(\xi) \end{aligned}$$

$$J_R(\xi) = \begin{bmatrix} J_R(\theta) & 0 \\ Q_R(\xi) & J_R(\theta) \end{bmatrix}_{6\times6} \qquad J_L(\xi) = \begin{bmatrix} J_L(\theta) & 0 \\ Q_L(\xi) & J_L(\theta) \end{bmatrix}_{6\times6}$$

Q_L : find out what's Q_L ?

$$\xi^\curlywedge = \begin{bmatrix} \theta^\wedge & 0 \\ \rho^\wedge & \theta^\wedge \end{bmatrix}_{6\times6} \qquad \begin{aligned} J_L(\xi) &\approx I_6 + \tfrac{1}{2}\xi^\curlywedge \\ J_R(\xi) &\approx I_6 - \tfrac{1}{2}\xi^\curlywedge \end{aligned}$$

How to define the error states of SE(3)

$$T = \begin{bmatrix} R & p \\ 0_{1\times3} & 1 \end{bmatrix} = \mathrm{Exp}\left(\begin{bmatrix} \theta \\ \rho \end{bmatrix}\right) = \begin{bmatrix} \mathrm{Exp}(\theta) & J_L(\theta)\,\rho \\ 0_{1\times3} & 1 \end{bmatrix}$$

$$T \cdot \delta T = \mathrm{Exp}(\lfloor {}^{0}_{\rho} \rfloor) \cdot \mathrm{Exp}(\lfloor {}^{\delta\theta}_{\delta\rho} \rfloor)$$

$$= \begin{bmatrix} R & p \\ 0_{1\times 3} & 1 \end{bmatrix} \begin{bmatrix} \mathrm{Exp}(\delta\theta) & J_L(\delta\theta)\,\delta\rho \\ 0_{1\times 3} & 1 \end{bmatrix}$$

$$= \begin{bmatrix} R & p \\ 0_{1\times 3} & 1 \end{bmatrix} \begin{bmatrix} \delta R & (I + \frac{1}{2}\lfloor \delta\theta \rfloor_\times)\,\delta\rho \\ 0_{1\times 3} & 1 \end{bmatrix}$$

$$= \begin{bmatrix} R & p \\ 0_{1\times 3} & 1 \end{bmatrix} \begin{bmatrix} \delta R & \delta\rho \\ 0_{1\times 3} & 1 \end{bmatrix} = \begin{bmatrix} R\delta R & R\delta\rho + p \\ 0_{1\times 3} & 1 \end{bmatrix}$$

Treat T as: $SO(3) + R^3$, Hence

$$T = \begin{bmatrix} R\delta R & p + \delta p \\ 0_{1\times 3} & 1 \end{bmatrix} = \begin{bmatrix} R\delta R & p + R R^T \delta p \\ 0_{1\times 3} & 1 \end{bmatrix} = \begin{bmatrix} R & p \\ 0_{1\times 3} & 1 \end{bmatrix} \begin{bmatrix} \delta R & R^T \delta p \\ 0_{1\times 3} & 1 \end{bmatrix}$$

Hence the difference between $SE(3)$ and $SO(3) + R^3$:

$$\begin{bmatrix} R & p \\ 0_{1\times 3} & 1 \end{bmatrix} \begin{bmatrix} \delta R & \delta\rho \\ 0_{1\times 3} & 1 \end{bmatrix} = \begin{bmatrix} R & p \\ 0_{1\times 3} & 1 \end{bmatrix} \begin{bmatrix} \delta R & R^T \delta p \\ 0_{1\times 3} & 1 \end{bmatrix}$$

$$\begin{cases} \delta\rho = R^T \delta p \\ \delta R = \delta R \end{cases}$$

The adjoint operation

For $SO(3)$: $R \cdot \mathrm{Exp}(\varphi) R^T = \mathrm{Exp}(R\varphi) \iff \mathrm{Ad}_R = R$

For $SE(3)$ $T\,\mathrm{Exp}(\xi) T^{-1} = \mathrm{Exp}(\mathrm{Ad}_T \cdot \varphi)$ Home work: prove it?

$$\mathrm{Ad}_T = \begin{bmatrix} R & 0 \\ \lfloor p \rfloor R & R \end{bmatrix}_{6\times 6}$$

Note that: Adjoint operation is a way to transfer the Lie algebra and Lie group

Ajoint is also Lie Group.

vector space $\longrightarrow$ Lie Algebra $\longrightarrow$ Lie Group

$$\xi = \begin{bmatrix} \theta \\ \rho \end{bmatrix}_{6\times1} \qquad ad_T = \xi^{\curlywedge} = \begin{bmatrix} \theta^\wedge & 0 \\ \rho^\wedge & \theta^\wedge \end{bmatrix}_{6\times6} \qquad Ad_T = \exp(\xi^{\curlywedge})$$

$$= \begin{bmatrix} R & 0 \\ (J_L(\theta)\rho)^\wedge R & R \end{bmatrix}_{6\times6}$$

Vector space $\longleftarrow$ Lie Algebra $\longleftarrow$ Lie Group

$$\log(Ad_T)^\vee = \begin{bmatrix} \theta \\ \rho \end{bmatrix} \longleftarrow \log(Ad_T) = \xi^{\curlywedge} = \begin{bmatrix} \theta^\wedge & 0 \\ \rho^\wedge & \theta^\wedge \end{bmatrix} \qquad Ad_T$$

A few important operations:

① Interpolation: $\quad {}^G_{t_1}T, \ {}^G_{t_2}T \qquad t \in [t_1, t_2]$

$${}^G_{t_1}T = \begin{bmatrix} {}^G_{t_1}R & {}^G p_{t_1} \\ 0_{1\times3} & 1 \end{bmatrix} \qquad {}^G_{t_2}T = \begin{bmatrix} {}^G_{t_2}R & {}^G p_{t_2} \\ 0_{1\times3} & 1 \end{bmatrix}$$

If SE(3), $T = T \cdot \mathrm{Exp}\left(\begin{bmatrix} \delta\theta \\ \delta\rho \end{bmatrix}\right)$, then

$${}^{t_1}_{t_2}\xi = \mathrm{Log}\left({}^G_{t_1}T^{-1} \cdot {}^G_{t_2}T\right) \qquad \lambda = \frac{t - t_1}{t_2 - t_1}$$

$${}^G_tT = {}^G_{t_1}T \cdot \mathrm{Exp}\left(\lambda \, {}^{t_1}_{t_2}\xi\right)$$

If $SO(3) + R^3$, $T = \begin{bmatrix} R\,\mathrm{Exp}(\delta\theta) & p + \delta p \\ 0_{1\times3} & 1 \end{bmatrix}$

$$\begin{cases} {}^G_tR = {}^G_{t_1}R \, \mathrm{Exp}\left(\lambda \log\left({}^G_{t_1}R^T \, {}^G_{t_2}R\right)\right) \\ {}^G p_t = {}^G p_{t_1} + \lambda\left({}^G p_{t_2} - {}^G p_{t_1}\right) \end{cases}$$

② compute $\dot{T}$.

If SE(3) $\quad T = T \cdot \mathrm{Exp}\left(\begin{bmatrix} \delta\theta \\ \delta\rho \end{bmatrix}\right)$

$$\frac{dT}{dt} = \lim_{\Delta t \to 0} \frac{T(t+\Delta t) - T(t)}{\Delta t} = \lim_{\Delta t \to 0} \frac{T \cdot \mathrm{Exp}(\lfloor \delta \rho^{\wedge} \rfloor) - T}{\Delta t}$$

$$= \lim_{\Delta t \to 0} \frac{\begin{bmatrix} R & p \\ 0_{1\times3} & 1 \end{bmatrix} \begin{bmatrix} \exp(\delta\theta) & J_l(\delta\theta)\delta\rho \\ 0_{1\times3} & 1 \end{bmatrix} - \begin{bmatrix} R & p \\ 0_{1\times3} & 1 \end{bmatrix}}{\Delta t}$$

$$= \lim_{\Delta t \to 0} \frac{\begin{bmatrix} R & p \\ 0_{1\times3} & 1 \end{bmatrix} \begin{bmatrix} \exp(\delta\theta) - I_3 & J_l(\delta\theta)\delta\rho \\ 0_{1\times3} & 0 \end{bmatrix}}{\Delta t}$$

$J_l(\delta\theta) \approx I + \frac{1}{2}\lfloor \delta\theta \rfloor_\times$

$$= \lim_{\Delta t \to 0} \frac{\begin{bmatrix} R & p \\ 0_{1\times3} & 1 \end{bmatrix} \begin{bmatrix} \lfloor \delta\theta \rfloor_\times & \delta\rho \\ 0_{1\times3} & 1 \end{bmatrix}}{\Delta t}$$

$$= \begin{bmatrix} R & p \\ 0_{1\times3} & 1 \end{bmatrix} \begin{bmatrix} \lfloor \omega \rfloor_\times & v \\ 0_{1\times3} & 0 \end{bmatrix}$$

v: is body velocity

$$= \begin{bmatrix} R\lfloor \omega \rfloor_\times & Rv \\ 0_{1\times3} & 0 \end{bmatrix}$$

If $SO(3) + R^3$:

$$\frac{dT}{dt} = \lim_{\Delta t \to 0} \frac{T(t+\Delta t) - T}{\Delta t}$$

$$= \lim_{\Delta t \to 0} \frac{\begin{bmatrix} R\,\mathrm{Exp}(d\theta) & p + \delta p \\ 0_{1\times3} & 1 \end{bmatrix} - \begin{bmatrix} R & p \\ 0_{1\times3} & 1 \end{bmatrix}}{\Delta t}$$

$$= \lim_{\Delta t \to 0} \frac{\begin{bmatrix} R\lfloor \delta\theta \rfloor_\times & \delta p \\ 0_{1\times3} & 0 \end{bmatrix}}{\Delta t}$$

$$= \begin{bmatrix} R\lfloor \omega \rfloor_\times & {}^G v \\ 0_{1\times3} & 0 \end{bmatrix}$$

${}^G v$ is the global velocity.

③ calibration: $\{I\}$, $\{C\}$ IMU frame and camera frame

Homework:

$${}^{C_1}_{C_2}T = {}^{C}_{I}T \cdot {}^{I_1}_{I_2}T \cdot {}^{C}_{I}T^{-1}$$

If $SE(3)$, ${}^{C_1}_{C_2}T = {}^{C}_{I}T \cdot {}^{I_1}_{I_2}T \cdot {}^{C}_{I}T^{-1}$

$$\mathrm{Exp}({}^{C_1}_{C_2}\xi) = \mathrm{Exp}(\mathrm{Adj}_{({}^{C}_{I}T)}\,{}^{I_1}_{I_2}\xi)$$

compute the Jacobians: $\dfrac{\partial\, {}^{C_1}_{C_2}\xi}{\partial\, {}^{I_1}_{I_2}\xi} = ?$ $\qquad \dfrac{\partial\, {}^{C_1}_{C_2}\xi}{\partial\, {}^{C}_{I}\xi} = ?$

$$\mathrm{Exp}({}^{C}_{I}\xi) = {}^{C}_{I}T$$

If $SO(3) + \mathbb{R}^3$, $\quad \dfrac{\partial \begin{bmatrix} {}^{C_1}_{C_2}\theta \\ {}^{C_1}p_{C_2} \end{bmatrix}}{\partial \begin{bmatrix} {}^{I_1}_{I_2}\theta \\ {}^{I_1}p_{I_2} \end{bmatrix}} = ? \qquad \dfrac{\partial \begin{bmatrix} {}^{C_1}_{C_2}\theta \\ {}^{C_1}p_{C_2} \end{bmatrix}}{\partial \begin{bmatrix} {}^{C}_{I}\theta \\ {}^{C}p_{I} \end{bmatrix}} = ?$

$SE_2(3)$

$$T = \begin{bmatrix} R & p & v \\ 0_{1\times3} & 1 & 0 \\ 0_{1\times3} & 0 & 1 \end{bmatrix}_{5\times5}$$

$$\xi = \begin{bmatrix} \theta \\ \rho \\ \upsilon \end{bmatrix} \to \xi^{\curlywedge} = \begin{bmatrix} \lfloor\theta\rfloor_\times & 0 & 0 \\ \lfloor\rho\rfloor_\times & \lfloor\theta\rfloor_\times & 0 \\ \lfloor\upsilon\rfloor_\times & 0 & \lfloor\theta\rfloor_\times \end{bmatrix}_{9\times9} \to T = \begin{bmatrix} R & J_l(\theta)\rho & J_l(\theta)\upsilon \\ 0_{1\times3} & 1 & 0 \\ 0_{1\times3} & 0 & 1 \end{bmatrix}_{5\times5}$$

The exponential operation:

$$\mathrm{Exp}(\xi) = \mathrm{Exp}\left(\begin{bmatrix} \theta \\ \rho \\ \upsilon \end{bmatrix}\right) = \begin{bmatrix} R & J_l(\theta)\rho & J_l(\theta)\upsilon \\ 0_{1\times3} & 1 & 0 \\ 0_{1\times3} & 0 & 1 \end{bmatrix}$$

The logarithmic operation:

$$\mathrm{Log}(T) = \mathrm{Log}\left(\begin{bmatrix} R & p & v \\ 0_{1\times3} & 1 & 0 \\ 0_{1\times3} & 0 & 1 \end{bmatrix}\right) = \begin{bmatrix} \theta \\ J_l^{-1}(\theta)\, p \\ J_l^{-1}(\theta)\, v \end{bmatrix}$$

The Adjoint operation:

$$\mathrm{Adj}_{T} = \begin{bmatrix} R & 0_3 & 0_3 \\ \lfloor p\rfloor_\times R & R & 0_3 \end{bmatrix}$$

$$
\begin{bmatrix} \lfloor p \rfloor_\times R & R & 0_3 \\ \lfloor v \rfloor_\times R & 0_3 & R \end{bmatrix}
$$

Finally Quaternion:

I put quaternion last, because I think SO(3), SE(3), Lie Group are good enough for future research.

Note that: There are majorly Hamilton and JPL quaternion.

They are generally similar except that Hamilton is right-handed while JPL is left handed.

For the future, I will use Hamilton quaternion notation.

$$
\begin{cases} Q = q_w + q_x i + q_y j + q_z k & \{1, i, j, k\} \quad i, j, k \text{ are imaginary axis} \\ i^2 = j^2 = k^2 = ijk = -1 & ij = jk = ki = 1 \end{cases}
$$

$$
\bar{q} = \begin{bmatrix} q_w \\ q_x \\ q_y \\ q_z \end{bmatrix} = \begin{bmatrix} q_w \\ q_v \end{bmatrix}
$$

For quaternion, the product is defined as:

$$
\bar{p} \otimes \bar{q} = \begin{bmatrix} p_w \\ p_v \end{bmatrix} \otimes \begin{bmatrix} q_w \\ q_v \end{bmatrix}
$$

$$
= \lfloor \bar{p} \rfloor_L \bar{q} \qquad \lfloor \cdot \rfloor_L \text{ left operation}
$$

$$
= \lfloor \bar{q} \rfloor_R \bar{p} \qquad \lfloor \cdot \rfloor_R \text{ right operation}
$$

$$\lfloor \bar{p} \rfloor_L = p_w I_4 + \begin{bmatrix} 0 & -p_v^T \\ p_v & \lfloor p_v \rfloor_\times \end{bmatrix}$$

$$\lfloor \bar{q} \rfloor_R = q_w I_4 + \begin{bmatrix} 0 & -q_v^T \\ q_v & -\lfloor q_v \rfloor_\times \end{bmatrix}$$

Identity: $\bar{q} = \begin{bmatrix} 1 \\ 0_{3\times1} \end{bmatrix}$

Unit quaternion:

$$\bar{q} = \begin{bmatrix} \cos\frac{\theta}{2} \\ \sin\frac{\theta}{2} \cdot k \end{bmatrix}$$

k: rotation axis

θ: rotation angle.

$R_{12} = R_1 \cdot R_2$

$\bar{q}_{12} = \bar{q}_1 \otimes \bar{q}_2$

Inverse: $\bar{q}^{-1} = \begin{bmatrix} \cos\frac{\theta}{2} \\ -\sin\frac{\theta}{2} k \end{bmatrix}$

why the quaternion is $\frac{\theta}{2}$?

$\{G\}$ Global frame, $\{I\}$: Imu frame.

$$^G\bar{p} = {}^G_I\bar{q} \otimes {}^I\bar{p} \otimes {}^G_I\bar{q}^{-1}$$

$$= \begin{bmatrix} q_w \\ q_v \end{bmatrix} \otimes \begin{bmatrix} 0 \\ ^Ip \end{bmatrix} \otimes \begin{bmatrix} q_w \\ -q_v \end{bmatrix}$$

$$= \begin{bmatrix} q_w \\ -q_v \end{bmatrix}_R \begin{bmatrix} q_w \\ q_v \end{bmatrix}_L \begin{bmatrix} 0 \\ ^Ip \end{bmatrix}$$

$$= \left(q_w I_4 + \begin{bmatrix} 0 & q_v^T \\ -q_v & \lfloor q_v \rfloor_\times \end{bmatrix}\right)\left(q_w I_4 + \begin{bmatrix} 0 & -q_v^T \\ q_v & \lfloor q_v \rfloor_\times \end{bmatrix}\right) \begin{bmatrix} 0 \\ ^Ip \end{bmatrix}$$

$$= \left(q_w^2 I_4 + 2q_w \begin{bmatrix} 0 & 0 \\ 0 & \lfloor q_v \rfloor_\times \end{bmatrix} + \begin{bmatrix} 0 & q_v^T \\ -q_v & \lfloor q_v \rfloor_\times \end{bmatrix} \begin{bmatrix} 0 & -q_v^T \\ q_v & \lfloor q_v \rfloor_\times \end{bmatrix}\right) \begin{pmatrix} 0 \\ ^Ip \end{pmatrix}$$

$$= \left(q_w^2 I_4 + 2q_w \begin{bmatrix} 0 & 0 \\ 0 & \lfloor q_v \rfloor_\times \end{bmatrix} + \begin{bmatrix} q_v^T q_v & 0 \\ \cdots & \cdots \end{bmatrix} \right) \begin{bmatrix} 0 \\ ^Ip \end{bmatrix}$$

$$\begin{bmatrix}0\\ {}^{G}p\end{bmatrix} = \begin{bmatrix} q_w^2 + q_v^T q_v & 0 \\ 0 & q_w^2 I_3 + 2q_w\lfloor q_v\rfloor_\times + q_v q_v^T + \lfloor q_v \rfloor_\times^2\end{bmatrix}\begin{bmatrix}0\\ {}^{I}p\end{bmatrix}$$

$${}^{G}p = \left(q_w^2 I_3 + 2q_w\lfloor q_v\rfloor_\times + q_v q_v^T + \lfloor q_v\rfloor_\times^2\right){}^{I}p$$

$$\lfloor q_v\rfloor_\times^2 = q_v q_v^T - q_v^T q_v I_3$$

$${}^{G}p = \left(I_3 + 2q_w\lfloor q_v\rfloor_\times + 2\lfloor q_v\rfloor_\times^2\right){}^{I}p$$

$$= \underbrace{\left(I_3 + \sin\theta\lfloor k\rfloor_\times + (1-\cos\theta)\lfloor k\rfloor_\times^2\right)}\,{}^{I}p$$

$$= {}^{C}_{I}R\;{}^{I}p$$

$$= \operatorname{Exp}(\theta k)\,{}^{I}p$$

Derivative

$${}^{G}_{t}\dot{\bar q} = \lim_{\Delta t\to 0}\frac{{}^{G}_{t+\Delta t}\bar q - {}^{G}_{t}\bar q}{\Delta t}$$

$$= \lim_{\Delta t\to 0}\frac{{}^{G}_{t}\bar q\otimes\begin{bmatrix}\cos\frac{\delta\theta}{2}\\ \sin\frac{\delta\theta}{2}k\end{bmatrix} - {}^{G}_{t}\bar q\otimes\begin{bmatrix}1\\0\end{bmatrix}}{\Delta t}$$

$$= \lim_{\Delta t\to 0}\frac{{}^{G}_{t}\bar q\otimes\begin{bmatrix}0\\ \frac{1}{2}\delta\boldsymbol{\theta}\end{bmatrix}}{\Delta t} \qquad \delta\boldsymbol{\theta} = \delta\theta\cdot k$$

$$= {}^{G}_{t}\bar q\otimes\begin{bmatrix}0\\ \frac{1}{2}\omega\end{bmatrix}$$

$$= \frac{1}{2}\left(0\cdot I_3 + \begin{bmatrix}0 & -\omega^T\\ \omega & -\lfloor\omega\rfloor_\times\end{bmatrix}\right){}^{G}_{t}\bar q$$

$$= \frac{1}{2}\begin{bmatrix}0 & -\omega^T\\ \omega & -\lfloor\omega\rfloor_\times\end{bmatrix}{}^{G}_{t}\bar q \qquad \Omega(\omega) \triangleq \begin{bmatrix}0 & -\omega^T\\ \omega & -\lfloor\omega\rfloor_\times\end{bmatrix}$$

$${}^{G}_{t}\dot{\bar q} = \frac{1}{2}\Omega(\omega)\,{}^{G}_{t}\bar q$$

Integration:

$$ {}^{G}_{t}\bar{q}, \quad w, \quad \delta t. \qquad \delta\boldsymbol{\theta} = w \cdot \delta t = \delta\theta \cdot k $$

$$ \delta\bar{q} = \begin{bmatrix} \cos\frac{\delta\theta}{2} \\ \sin\frac{\delta\theta}{2} \cdot k \end{bmatrix} \qquad {}^{G}_{t+\delta t}\bar{q} = {}^{G}_{t}\bar{q} \otimes \delta\bar{q} $$